Lecture 6: Residential Mortgages and Mortgage-Backed Securities

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Learning Objectives

By the end of this lecture you should be able to:

- Explain how residential mortgage loans are originated and understand underwriting standards such as payment-to-income (PTI) and loan-to-value (LTV) ratios.
- Distinguish among different types of residential mortgage loans according to lien status, credit classification, interest rate type, amortization structure, credit guarantees, loan size and the presence of prepayment penalties.
- Describe the characteristics of conforming loans and understand why they matter for securitization.
- Identify the risks associated with investing in mortgage loans, including credit risk, liquidity risk, price risk and prepayment/cash-flow uncertainty.
- Recognise the major sectors of the residential mortgage-backed security (RMBS) market and describe agency pass-through securities.
- Explain prepayment conventions such as Conditional Prepayment Rate (CPR), Single Monthly Mortality (SMM) and Public Securities Association (PSA) benchmarks, and construct monthly cash flows from these measures.
- Discuss factors affecting prepayments and the modelling of prepayment behaviour.
- Compute cash-flow yield and bond-equivalent yield for mortgage pools and understand their limitations.

- Describe the role of average life, yield spreads to Treasuries and prepayment risk in asset/liability management for mortgage investors.
- Understand how mortgage-backed securities trade in the secondary market.

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1 Origination of Residential Mortgage Loans

Mortgage origination refers to the process by which lenders (often banks or mortgage companies) approve and fund loans that allow individuals to buy homes. The underwriting process determines whether a borrower meets the lender's standards.

1.1 Underwriting Standards

Underwriters examine the borrower's ability and willingness to repay the loan. Two key ratios are used:

- **Payment-to-Income (PTI) ratio** – The PTI compares a borrower's monthly housing payment to their monthly income. Lenders typically require a PTI ratio below 28–36%, meaning that no more than one-third of income is devoted to housing expenses.
- **Loan-to-Value (LTV) ratio** – The LTV is the loan amount divided by the appraised value of the property. A lower LTV indicates more borrower equity, reducing credit risk. Conventional mortgages often require an LTV of 80% or lower, implying a down payment of at least 20%.

1.2 Documentation and Credit Scores

Underwriters also verify employment history, credit scores and assets. High credit scores allow borrowers to qualify for lower interest rates. Documents such as pay stubs, W-2 forms and tax returns verify income; bank statements verify assets for down payment and reserves.

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2 Types of Residential Mortgage Loans

Mortgage loans vary widely. Below we classify loans by key features.

2.1 Lien Status

A **first-lien** mortgage has priority over other claims on the property; if the borrower defaults, the first-lien holder receives proceeds from foreclosure before junior liens. **Second liens** or **home equity loans** carry higher risk and generally higher interest rates.

2.2 Credit Classification

Borrowers are often grouped by credit quality:

- **Prime mortgages**: borrowers have high credit scores, stable income and low debt, resulting in lower default risk.
- **Alt-A**: credit quality is between prime and subprime; borrowers may have limited documentation or higher LTVs but still relatively good credit histories.
- **Subprime mortgages**: borrowers have lower credit scores and higher default risk; these loans typically have higher interest rates and may feature structures such as adjustable rates or significant fees.

2.3 Interest Rate Type

- **Fixed-rate mortgages (FRMs)** have constant interest rates for the life of the loan, making monthly payments predictable. A 30-year FRM is a common example.
- **Adjustable-rate mortgages (ARMs)** have rates that reset periodically based on a reference index plus a margin. For instance, a 5/1 ARM has a fixed rate for five years and adjusts annually thereafter.

2.4 Amortization Type

- **Fully amortizing loans** require payments that cover both principal and interest so that the loan balance declines to zero by maturity.
- **Interest-only loans** require only interest payments for an initial period; principal payments start later or at maturity, increasing repayment risk.
- **Balloon mortgages** amortize partially but require a large lump-sum payment at the end; they have refinance risk.

2.5 Credit Guarantees and Insurance

Some loans carry **government guarantees**. For example, Federal Housing Administration (FHA) loans are insured by the U.S. government, and Veterans Affairs (VA) loans are guaranteed for qualified military borrowers. These loans enable higher LTVs with lower down payments.

2.6 Loan Balances

Loans are described as **conforming** or **jumbo** based on size. Conforming loans must meet agency guidelines, including a maximum loan amount. Jumbo loans exceed these limits and typically carry higher rates due to lower liquidity and lack of government guarantee.

2.7 Prepayments and Prepayment Penalties

Borrowers often have the right to **prepay** part or all of the mortgage balance without penalty. Lenders sometimes charge prepayment penalties to deter early refinancing and protect interest income. Penalties may be a percentage of the outstanding balance if the loan is paid off within a specified period.

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3 Conforming Loans

Conforming loans meet the underwriting guidelines of the government-sponsored enterprises (GSEs) Fannie Mae and Freddie Mac. Key criteria include:

- Loan balance below the conforming loan limit (which changes annually and varies by region).
- Borrower credit score above a minimum threshold (often 620+).
- Maximum LTV ratio, typically 80% (higher with mortgage insurance).
- Documentation of income and assets.

Because conforming loans can be packaged into agency mortgage-backed securities, they generally have lower interest rates and greater liquidity than non-conforming or jumbo loans.

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4 Risks Associated with Investing in Mortgage Loans

Investing in mortgages involves several sources of risk.

- **Credit risk**: the risk that borrowers will default. Higher-LTV and lower-credit-score loans have more credit risk. Government insurance and credit enhancement reduce this risk.

- **Liquidity risk**: mortgages are less liquid than Treasury securities. Secondary market depth varies by loan type; large institutional investors may sell pools or securitised mortgages more easily than individual whole loans.
- **Price risk**: the market value of mortgages fluctuates with interest rates. Because mortgages contain prepayment options, their effective duration changes as rates move, making hedging complex.
- **Prepayment and cash-flow uncertainty**: borrowers may prepay their mortgages whenever interest rates fall or life events prompt relocation. Investors therefore face reinvestment risk when prepayments occur and extension risk when prepayments slow.

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5 Sectors of the Residential Mortgage-Backed Security Market

Residential mortgage loans are often pooled and sold as securities. The main sectors include:

- **Agency pass-through securities**: pools of conforming mortgages guaranteed by GSEs (Fannie Mae, Freddie Mac) or the Government National Mortgage Association (GNMA). Investors receive pro-rata shares of principal and interest cash flows, with the guarantee of timely payments (GNMA) or guarantee of principal (Fannie/Freddie).
- **Agency collateralized mortgage obligations (CMOs)**: multi-tranche structures that redirect principal and prepayment risk among various classes to meet different investor needs.
- **Non-agency RMBS**: securities backed by non-conforming or subprime mortgages without GSE guarantees. Credit enhancements such as senior/subordinate tranches or reserve funds mitigate default risk.
- **Whole-loan trading**: transactions of pools of mortgages that are not securitized. Banks and insurance companies often hold whole loans.

The agency sector dominates in size because of the GSEs' guarantee. Non-agency and private-label securities became significant during periods of active securitization (e.g., mid-2000s) but are more vulnerable to credit risk.

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6 General Description of an Agency Mortgage

Agency mortgages are loans that conform to GSE guidelines and are either guaranteed or insured. The loan pools are assembled by originators and sold to GSEs or securitised into pass-throughs.

Investors in agency pass-throughs face prepayment risk but are protected from credit losses through the GSE guarantee. Cash flows consist of scheduled principal and interest plus unscheduled prepayments.

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7 Issuers of Agency Pass-Through Securities

- **Fannie Mae** (Federal National Mortgage Association) – purchases mortgages from lenders and issues pass-through securities; guarantees timely payment of principal and interest.
- **Freddie Mac** (Federal Home Loan Mortgage Corporation) – similar role; historically focused on savings institutions and thrifts.
- **GNMA (Ginnie Mae)** – a government agency that guarantees mortgage pools composed of FHA/VA loans. GNMA securities carry the full faith and credit of the U.S. government and guarantee timely payment of principal and interest.

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8 Prepayment Conventions and Cash Flow

Accurately projecting the cash flows of mortgage pools requires assumptions about prepayment behaviour. Several conventions are used.

8.1 Conditional Prepayment Rate (CPR)

The CPR is an annualised rate expressing the percentage of the outstanding principal that is expected to be prepaid in a year. For example, a CPR of 6% implies that 6% of the mortgage pool's remaining principal will be prepaid over the next year.

8.2 Single Monthly Mortality (SMM) Rate

The SMM converts the annual CPR into a monthly probability of prepayment:

$$\text{SMM} = 1 - (1 - \text{CPR})^{1/12}.$$

If CPR = 6%, then (SMM 0.516%). This means that approximately 0.516% of the outstanding principal will prepay each month.

8.3 SMM and Monthly Prepayment

The expected prepayment for month t is:

$$\text{Prepayment}_t = \text{SMM} \times (\text{Beginning balance}_t - \text{Scheduled principal}_t).$$

Scheduled principal is the amount of principal included in the regular monthly mortgage payment.

8.4 Public Securities Association (PSA) Benchmark

The PSA benchmark is a prepayment standard used to model mortgage pools. Under **100% PSA**, prepayments start at 0.2% CPR in the first month and increase by 0.2% each month until reaching 6% CPR at month 30, after which the CPR remains at 6%.

Higher PSA speeds (e.g., 150% PSA) scale the benchmark by that factor. Thus 150% PSA means the prepayment curve ramps up 50% faster than the benchmark.

8.5 Monthly Cash Flow Construction

For each month, the cash flow consists of:

1. **Interest**: computed on the previous month's balance.
2. **Scheduled principal**: the portion of the payment allocated to principal.
3. **Prepayment**: derived from SMM.

The sum of scheduled principal and prepayment gives the total principal paydown. The remaining balance is reduced accordingly.

8.6 Beware of Conventions

While CPR, SMM and PSA provide useful frameworks, actual prepayments depend on many factors, and models may misestimate future behaviour. Analysts should examine historical data, current economic conditions and borrower incentives when choosing a prepayment assumption.

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9 Factors Affecting Prepayments and Prepayment Modelling

Prepayments are influenced by borrower behaviour and interest rates. Key components include:

9.1 Housing Turnover Component

Borrowers may prepay when they sell their homes. Housing turnover rates are influenced by labour mobility, household life events (marriage, childbirth, divorce) and regional economic conditions.

9.2 Cash-Out Refinancing Component

When home values rise, borrowers may refinance to extract equity for consumption or investment. Cash-out refinancing is more common when housing prices are appreciating rapidly.

9.3 Rate/Term Refinancing Component

Borrowers refinance to obtain lower interest rates or shorten/lengthen the term of their mortgages. The incentive to refinance is often measured by the difference between the borrower's current mortgage rate and prevailing rates minus the costs of refinancing.

9.4 Modelling Approaches

Prepayment models typically separate the above components and include variables such as current and lagged mortgage rates, borrower equity and macroeconomic factors. Large institutions use econometric models or option-adjusted spread (OAS) frameworks to value mortgage instruments under different interest rate scenarios.

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10 Cash Flow Yield and Related Measures

10.1 Cash Flow Yield

The **cash flow yield** (CFY) is the discount rate that equates the present value of expected cash flows of a mortgage pool to its market price. Because prepayments shorten the effective maturity, the CFY is often higher than the mortgage's coupon if prepayment speeds are rapid.

10.2 Bond-Equivalent Yield

To compare mortgage yields with Treasury yields, analysts convert the CFY to a **bond-equivalent yield** (BEY) by doubling the semi-annual yield. BEY $= 2 \times$ (semi-annual yield).

10.3 Limitations of Cash Flow Yield Measure

The CFY assumes the assumed prepayment pattern will occur exactly as forecast. If actual prepayments differ, the realised yield will deviate from CFY. Mortgage-backed securities therefore often trade on the basis of **option-adjusted spread** (OAS), which accounts for interest rate volatility and prepayment option value.

10.4 Yield Spread to Treasuries and Average Life

Investors compare mortgage pools to risk-free Treasuries by quoting a **spread** over the Treasury curve. **Average life** is the weighted average time to receipt of principal payments and reflects how prepayments shorten the effective maturity. Two pools with the same average life can have different prepayment profiles.

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11 Prepayment Risk and Asset/Liability Management

Mortgage investors such as banks and insurance companies face **prepayment risk**: the risk that mortgages will be repaid earlier (forcing reinvestment at lower rates) or later (exposing investors to higher market yields). Institutions managing liabilities (e.g., savings accounts) must match asset cash flows to liability outflows.

Asset/liability management (ALM) strategies involve:

- Adjusting the composition of mortgage assets to target a desired average life and duration.
- Using derivatives (e.g., interest rate swaps, swaptions) to hedge interest rate exposure and prepayment risk.
- Diversifying across sectors (agency vs non-agency) and coupon levels to manage convexity risk.

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12 Secondary Market Trading

The secondary market for mortgages includes:

- **Whole loan trading**: banks and institutional investors buy and sell pools of loans. Transactions are negotiated bilaterally or through brokers.
- **Agency pass-through trading**: GSE-guaranteed pools trade over the counter with dealers quoting prices in 32nds. Settlement occurs monthly.
- **Mortgage-backed derivatives**: CMOs, stripped mortgage securities and interest-only/principal-only (IO/PO) securities trade to allow investors to tailor prepayment exposure.
- **TBA (To Be Announced) market**: standardized forward contracts for agency pass-throughs. Buyers and sellers agree on general characteristics (coupon, maturity, agency) but not the specific pool; the actual pool is delivered at settlement. The TBA market provides liquidity comparable to Treasuries and is essential for mortgage rate locks.

Conclusion

Residential mortgage finance underpins a major portion of the fixed-income market. Understanding underwriting standards, loan classifications and prepayment behaviour is crucial for valuing and managing mortgage investments. Mortgage-backed securities introduce complexity because of embedded prepayment options, but they also provide diversified income streams, potential yield premiums over Treasuries and tools for asset/liability management. Investors must carefully assess prepayment risk, credit quality and liquidity when allocating capital to this sector.